After the coupled signal reaches the users premises, another modem splits the pair; the cable channels head to your television and data from the Internet makes its way to your computer.

A segment of about 6 MHz is allotted to the Internet data, and such a segment is termed as a channel. This fixed capacity is enough to support up to a thousand subscribers. Problems arise when the number exceeds that mark. Aniruddha Gupte, a broadband hopeful, had subscribed to just such a service (7 Star Cable Network, which has leased lines from the ISP, NetMagic). He was promised browsing speeds around the 48 Kbps mark—not a true broadband connection for sure, but Gupte took solace in the fact that he would not have to pay monthly telephone bills and would be granted an always-on, dedicated line to the Net. He paid Rs 1,000 as installation charges and then dished out monthly rental charges that came up to an equal amount.

> "I am ready to switch back to dial-up: talk about backward compatibility!"
>
> *—Aniruddha Gupte, bemoaning the fact that his broadband connection is slower than a dial-up line*

As is financially feasible, Cable modems are only seen at select installations and most of the throughput is via switches and hubs installed on the terraces of buildings. Gupte found that he was connected in a similar way: a hub sits on the terrace of his building, connecting his and five other houses via a Cat-5 cable. The promised speed of 48 Kbps did make a brief appearance, but as new users were added, the speed quickly degraded and things reached such a stage that hitting a measly 20 Kbps mark (a dial-up connection offers twice as much speed!) was looked upon as miraculous. Needless to say, Gupte is an unhappy surfer. "The connection is worse than a dial-up line. The bandwidth is so bad that one site takes 3 to 4 minutes to load!" To further elucidate the fact that the bandwidth is being choked by the number of subscribers, Gupte presents this case: "Like everyone else, I like to chat on MSN when I am online. Sadly, just surfing the Internet takes up so much of the juice available that every time I visit a Web page, MSN thinks that the connection is dead and promptly signs off!" To ease the saturation of the allotted 6 MHz bandwidth, the service provider can add another channel and split the user base along the two. It is a solution that requires additional monetary investment. Gupte did complain to the concerned party but till date no action has been taken. This is a problem that any Cable modem based service will eventually experience. DSL lines also face inherent problems. When you are closer to the provider's central office, the connection is faster for receiving data than it is for sending data over the Internet and is generally the costliest solution in the offing.

So you Think you Have Broadband?

1. **How long does it take you to access a Web site?**
   a. I just need to click my heels and the Web page loads
   b. A quick, two-minute noodle session and I am off surfing
   c. I always find the time to do my laundry in between Web pages

2. **How long does it take you to download a music file from the Internet?**
   a. You shall be mine post-lunch, O MP3 file!
   b. I have to keep the computer on overnight for the file to download
   c. Hey, I get the latest music albums before they are officially released

3. **When was the last time you enjoyed a great action game online?**
   a. Does a card game count?
   b. My ping rate is longer than my PIN code
   c. They call me a low-ping b******!

4. **Do you frequently chat with your girlfriend in the US using a Web cam?**
   a. Yeah, but her mom keeps popping up (spoilsport!)
   b. Yes, but she lives across the street
   c. Hey, I'm grateful I can chat uninterrupted

5. **Quicktime.com, quick what pops into your head?**
   a. Ewww, Apple! (The grapes are sour, the grapes are sour, the…)
   b. Movies, masti, magic!
   c. Why won't they let me download those trailers?

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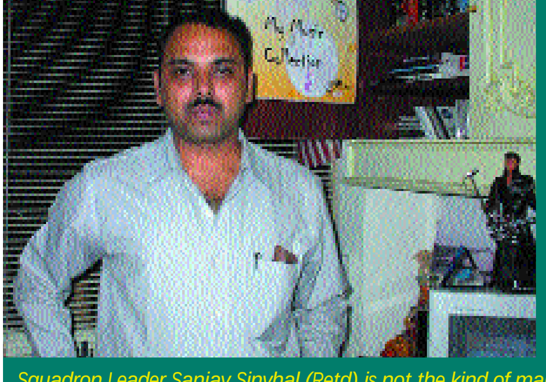

Squadron Leader Sanjay Sinvhal (Retd) is not the kind of man to shy away from a fight. So when he thought that Powersurfer was being unkind to its customers, he promptly took up the cudgels on behalf of the aggrieved masses

Barbed wire

As if these intrinsic walls were not enough to limit the user's experience, some providers put up barbed fences in the form of bandwidth caps. Such a cap is an artificial limit on the maximum speed a customer can achieve. This is done by monitoring their current usage and throttling network packets as and when necessary.